Introduction to Actuarial Audit under IFRS 17

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Agenda

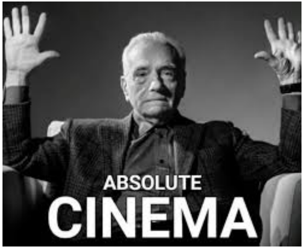
My Personal Goal from Today's Session I

1. Sharing 30% accounting knowledge 70% actuarial application
2. Showing you some memes along the presentation so keep your eyes open :D
3. Audit is about how you think, not just what you do.
4. Prevent one of these things to happen to you during your first internship/job

(Thanks to: John Lee)

My Personal Goal from Today's Session II

5. Your natural reaction after the sharing session:



Out of Scope

1. Reinsurance Contract Held
2. Shariah Insurance
3. Goodwill
4. Insurance Merger and Acquisition (M&A)
5. Making you an IFRS17 wizard

Meme I have to put in but don't think fit anywhere in the topic

soon

Disclaimer

Please read

- **Purpose:** knowledge sharing / general educational content.
- **Not advice:** this is *not* actuarial, accounting, audit, or legal advice.
- **Use at your own risk:** any reliance or decisions are the user's responsibility.
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Actual Science and Accounting

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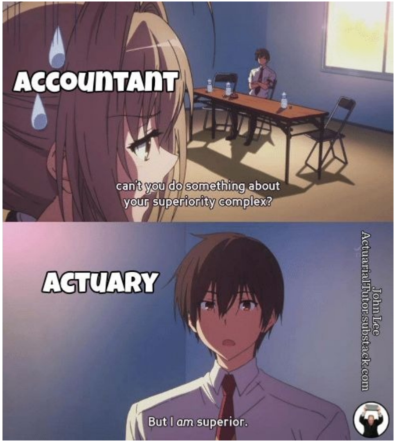
Actuaries and Accountants



Why do these memes exist? Is it true? (Credit to:John Lee)

Accounting Standars for Insurance

In (re)insurance one of the International Financial Reporting Standards (IFRS) we use is IFRS 17 : Insurance Contract. In every IFRS, it has 3 major sections:

1. Scope: which items are covered by the IFRS (accounting),
2. Recognition and Measurement: how to record the initial balance (actuarial science),
3. Disclosure: how to present the result in financial disclosures (accounting).

Doing the actuarial audit, means that we need to check the **documentation, assumptions used** and whether it align with the **standards**.

Example of IFRS17 Financial Statement - Balance Sheet - Life Insurance

Below is the Prudential Life snapshots of IFRS17 disclosure access on 6 February 2026.

	Note	2025 $m 30 Jun	2024 $m 30 Jun	2024 $m 31 Dec
Assets				
Goodwill	C4.1	**889**	819	848
Other intangible assets	C4.2	**3,939**	3,758	3,824
Property, plant and equipment	C1.2	**537**	390	417
Insurance contract assets	C3.1	**1,722**	1,131	1,345
Reinsurance contract assets	C3.1	**3,267**	3,200	3,390
Deferred tax assets		**150**	155	142
Current tax recoverable		**75**	25	31
Investments in joint ventures and associates accounted for using the equity method		**2,517**	1,781	2,412
Investment properties	C1.1	**3**	3	3
Loans	C1.1	**534**	543	517
Equity securities and holdings in collective investment schemes note	C1.1	**83,705**	73,110	81,002
Debt securities note	C1.1	**84,871**	74,543	73,804
Derivative assets	C1.1	**1,528**	276	395
Deposits	C1.1	**6,141**	5,284	5,466
Accrued investment income	C1.2	**1,039**	960	902
Other debtors	C1.2	**2,284**	2,440	1,310
Assets held for sale	C1.2	**282**	291	296
Cash and cash equivalents	C1.1	**5,636**	5,978	5,772
Total assets		**199,119**	174,687	181,876

	Note	2025 $m 30 Jun	2024 $m 30 Jun	2024 $m 31 Dec
Equity				
Shareholders' equity	C3.1	**18,119**	16,171	17,492
Non-controlling interests		**1,325**	1,065	1,182
Total equity		**19,444**	17,236	18,674
Liabilities				
Insurance contract liabilities	C3.1	**161,476**	141,099	147,566
Reinsurance contract liabilities	C3.1	**510**	1,379	536
Investment contract liabilities without discretionary participation features	C2.2	**730**	819	748
Core structural borrowings of shareholder-financed businesses	C5.1	**4,473**	3,930	3,925
Operational borrowings	C5.2	**853**	961	797
Obligations under funding, securities lending and sale and repurchase agreements		**665**	576	272
Net asset value attributable to unit holders of consolidated investment funds		**2,332**	2,921	2,679
Deferred tax liabilities		**1,772**	1,339	1,514
Current tax liabilities		**273**	231	238
Accruals, deferred income and other creditors	C1.2	**5,235**	3,395	2,848
Provisions		**157**	137	218
Derivative liabilities		**924**	426	1,617
Liabilities held for sale	C1.2	**275**	238	244
Total liabilities		**179,675**	157,451	163,202
Total equity and liabilities		**199,119**	174,687	181,876

Note
Included within equity securities and holdings in collective investment schemes and debt securities as at 30 June 2025 are $1,599 million of lent securities and assets subject to repurchase agreements (30 June 2024: $1,680 million; 31 December 2024: $1,565 million).

Observe that the Insurance Contract Liabilities (ICL) is around 80% of the total equity and liabilities, which make it more important to verify whether the amount is correct.

Example of IFRS17 Financial Statement - Profit and Loss - Life Insurance

	Note	2025 $m Half year	2024 $m Half year	2024 $m Full year
Insurance revenue	B1.4	**5,326**	4,961	10,358
Insurance service expenses		**(3,961)**	(3,638)	(7,763)
Net expense from reinsurance contracts held		**(125)**	(252)	(302)
Insurance service result		**1,240**	1,071	2,293
Investment return	B1.4	**7,059**	2,495	5,919
Fair value movements on investment contract liabilities		**(14)**	(54)	(95)
Net insurance and reinsurance finance expense		**(6,149)**	(2,274)	(4,492)
Net investment result		**896**	167	1,332
Other revenue	B1.4	**189**	197	382
Non-insurance expenditure		**(487)**	(532)	(1,003)
Finance costs: interest on core structural borrowings of shareholder-financed businesses		**(87)**	(85)	(171)
Loss attaching to corporate transactions	B1.1	**(16)**	(69)	(71)
Share of (loss) profit from joint ventures and associates, net of related tax		**(28)**	(243)	477
Profit before tax *(being tax attributable to shareholders' and policyholders' returns)* note		**1,707**	506	3,239
Tax charge attributable to policyholders' returns		**(7)**	(112)	(286)
Profit before tax attributable to shareholders' returns	B1.1	**1,700**	394	2,953
Total tax charge attributable to shareholders' and policyholders' returns	B2	**(348)**	(324)	(824)
Remove tax charge attributable to policyholders' returns		**7**	112	286
Tax charge attributable to shareholders' returns		**(341)**	(212)	(538)
Profit for the period		**1,359**	182	2,415

Top 4 of the largest line items are investment return, net insurance and reinsurance finance epxense insurance revenue and insurance service expenses

Example of IFRS17 Financial Statement - Balance Sheet - General Insurance

Below is the Zurich Insurance snapshots of IFRS17 disclosure access on 9 February 2026.

Consolidated balance sheets

Assets

in USD millions, as of December 31	Notes	2024	2023
Assets			
Cash and cash equivalents		**6,768**	**7,280**
Total Group investments	5	152,562	140,966
Equity securities		14,182	13,217
Debt securities		118,415	105,924
Investment property		11,734	13,684
Mortgage loans		4,047	4,324
Other financial assets		4,039	3,682
Investments in associates and joint ventures		146	135
Investments for unit-linked contracts		148,535	141,144
Total investments		**301,098**	**282,110**
Insurance contract assets	7	768	580
Reinsurance contract assets	7	21,450	21,942
Receivables and other assets	14	11,717	10,391
Deferred tax assets	16	1,703	1,700
Assets held for sale[1]	4	1,203	23,758
Property and equipment	12	1,867	2,092
Attorney-in-fact contracts	13	2,650	2,650
Goodwill	13	4,805	4,541
Other intangible assets	13	3,977	4,337
Total assets		**358,005**	**361,382**

1 As of December 31, 2024, the Group had USD 1.2 billion of assets held for sale based on agreements signed to sell portfolios of Zurich Insurance plc (now known as Zurich Insurance Europe AG) and Zurich Insurance Company Ltd, UK Branch (see note 4). In 2023, the Group had USD 23.8 billion of assets held for sale based on agreements signed to sell portfolios of Zurich Deutscher Herold LebensversicherungAktiengesellschaft, Zurich Chile Seguros de Vida S.A., Zurich Insurance plc (now known as Zurich Insurance Europe AG) and Zurich Insurance Company Ltd, UK Branch (see note 4).

Liabilities and equity

in USD millions, as of December 31	Notes	2024	2023
Liabilities			
Liabilities for investment contracts	8	66,507	60,270
Insurance contract liabilities	7	230,479	216,962
Reinsurance contract liabilities	7	437	504
Obligation to repurchase securities		1,123	796
Other liabilities[1]	15	16,022	16,661
Deferred tax liabilities	16	2,446	2,300
Liabilities held for sale[2]	4	1,162	23,860
Senior debt	17	4,020	5,190
Subordinated debt	17	8,871	8,559
Total liabilities		**331,067**	**335,102**
Equity			
Share capital	18	10	10
Additional paid-in capital	18	1,410	1,333
Net unreal. gains/(losses) on financial assets		(4,452)	(4,307)
Change in discount rate for (re)insurance contract		4,372	4,291
Change in fair value of underlying items		1,158	1,053
Cumulative foreign currency translation adjustment		(11,103)	(10,616)
Revaluation reserves		254	254
Retained earnings		33,823	32,842
Shareholders' equity		25,472	24,860
Non-controlling interests		1,466	1,419
Total equity		**26,938**	**26,280**
Total liabilities and equity		**358,005**	**361,382**

1 Includes restructuring provisions, litigation and regulatory provisions (see note 15) and other provisions.
2 As of December 31, 2024, the Group had USD 1.2 billion of liabilities held for sale based on agreements signed sell portfolios of Zurich Insurance plc (now known as Zurich Insurance Europe AG) and Zurich Insurance Company Ltd, UK Branch (see note 4). In 2023, the Group had USD 23.9 billion of liabilities held for sale based on agreements signed to sell portfolios of Zurich Deutscher Herold LebensversicherungAktiengesellschaft, Zurich Chile Seguros de Vida S.A.,Zurich Insurance plc (now known as Zurich Insurance Europe AG) and Zurich Insurance Company Ltd, UK Branch (see note 4).

Observe that the Insurance Contract Liabilities (ICL) is around 60% of the total equity and liabilities, which make it more important to verify whether the amount is correct.

Example of IFRS17 Financial Statement - Profit and Loss - General Insurance

Consolidated income statements

in USD millions, for the years ended December 31	Notes	2024	2023
Insurance revenue	9	59,507	56,099
Insurance service expense		(50,456)	(47,422)
Net expenses from reinsurance contracts held		(3,031)	(2,981)
Insurance service result		6,020	5,696
Net investment income on Group investments		5,730	5,382
Net capital gains/(losses) on Group investments		1,084	(696)
Net investment result on Group investments	5	6,814	4,687
Net investment result on unit-linked investments		16,384	14,191
Change in liabilities for investment contracts and other funds		(8,112)	(6,378)
Re-/insurance finance income/(expenses)		(12,244)	(10,963)
Net investment result		2,842	1,536
Fee income	10	6,011	5,885
Fee business expenses	10	(3,575)	(3,583)
Fee result		2,436	2,303
Other revenues		358	210
Net gains/(losses) on divestment of businesses	4	114	(104)
Interest expense on debt		(440)	(456)
Other expenses	11	(2,864)	(2,727)
Other result		(2,832)	(3,077)
Net income before income taxes		8,465	6,458
of which: Attributable to non-controlling interests		566	536
Income tax (expense)/benefit	16	(2,261)	(1,741)
attributable to policyholders		(179)	(172)
attributable to shareholders		(2,082)	(1,568)
of which: Attributable to non-controlling interests		(176)	(170)
Net income after taxes		**6,204**	**4,717**
attributable to non-controlling interests		390	366
attributable to shareholders		5,814	4,351

Overview of IFRS 17

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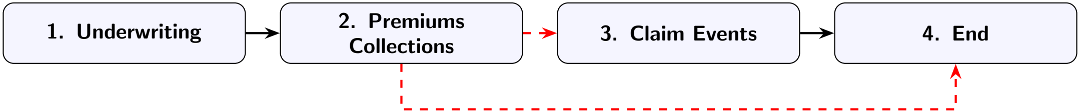
Introduction to Insurance Contract

Insurance contract is a contract under which one party (the issuer) accepts **significant insurance risk** from another party (the policyholder) by agreeing to **compensate** the policyholder if a specified **uncertain future event** (the insured event) adversely affects the policyholder.

Source: IFRS 17, Appendix A (Definitions).

Typical lifecycle of an insurance as a policyholder:



But how to determine the **significant insurance risk**?

What is an insurance contract (conceptually)? I

Insurance risk is significant if and only if, an insured event could cause the issuer to pay **additional amounts** that are significant in any single scenario, excluding scenarios that have no commercial substance (i.e., no discernible effect on the economics of the transaction)...
Source: IFRS 17, Appendix B18

The additional amounts above are determined on a present-value basis.
Source: IFRS 17, Appendix B20

What is an insurance contract (conceptually)? II

The additional ammounts refer to the present value of amounts that exceed those that would be payable if no insured event had occurred (excluding scenarios that lack commercial substance). Those additional amounts include claims handling and assessment costs, but exclude:

- the loss of the ability to charge the policyholder for future service,
- a waiver, on death, of charges that would be made on cancellation or surrender,
- a payment conditional on an event that does not cause a significant loss to the holder of the contract,
- possible reinsurance recoveries.

Source: IFRS 17, Appendix B21

Or as an actuary, this means to:

$$\frac{\sum_i E\left[\mathrm{PVCF}_{i,\text{ claim event}}\right]}{\sum_i E\left[\mathrm{PVCF}_{i,\text{ no claim event}}\right]} \tag{1}$$

Quick Break

From this section, now you know how to answer this question from a very famous *"star"*

Recognition and Measurement under IFRS 17

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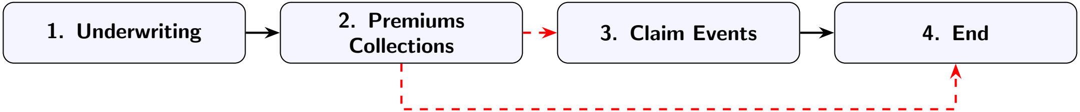
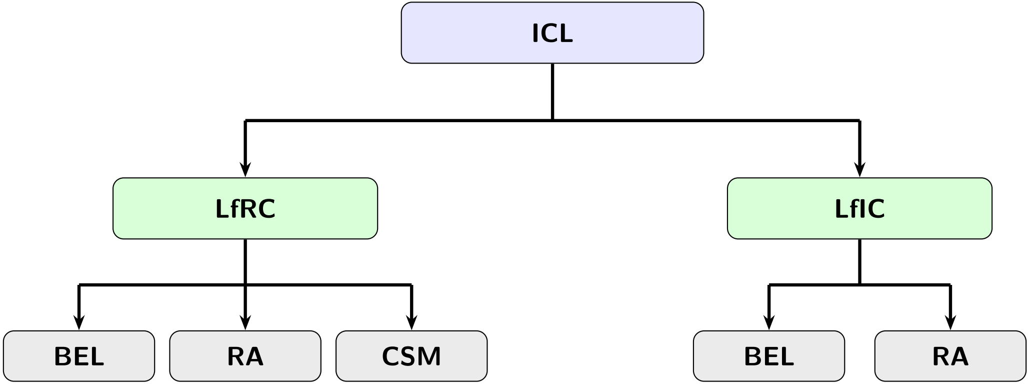
Where's My Money?



We are going to take a closer look on the lifecyle of an insurance from previous slide and ask ourselves, "what happened to the money after policyholders pay the premium? Does the insurer just keep the money in a bank? Pay claims to other policyholder? or just hope that no claim will submitted ?

When the insurance issued, the insurance directly become a liability to the insurrer meaning that the insurance company is liable to pay the policy holder if a claim event happened. But it doesn't mean that every insurance will be a libility to the insurance company. There will be some cases where the insurance can be an assets (as we see in the previous balance sheet examples).

What's inside the Insurance Contract Liability (ICL)?



Level of Aggregation (LoA)

Before we move to the building blocks, and measurement model, IFRS17 requires to aggregate (group) insurance contracts into portfolios. IFRS 17 is clear on how contracts should be grouped. There are three criteria:

- Similar risk: product type or characteristics (e.g., premium payment mode),
- Profitability: expected profitability of the product,
- Annual cohort: issue year of the insurance contract. IFRS17 prohibits to include contracts issued more than one year apart in the same group

Source: IFRS 17, Para. 14, 16, 22

This aggregation is done only at the initial recognition of the contract *Source: IFRS 17, Para. 24*

Level of Aggregation (LoA) - Profitability I

IFRS17 requires that the product is divided at least into 3 profitability group i.e.,:

- Onerous at initial recognition
- No Significant Possibility of Becoming Onerous (NSPBO)
- Remaining

Source: IFRS 17, Para. 16

IFRS17 defines onerous insurance contract at the date of initial recognition if the **Fulfilment Cash Flows (FCF)** allocated to the contract, any previously recognised **Insurance Acquisition Cash Flows (IACF)** and any cash flows arising from the contract at the date of initial recognition in total are a **net outflow**.

Source: IFRS 17, Para. 47

Level of Aggregation (LoA) - Profitability II

There is no specific guideline from IFRS 17 in determining NSPBO and remaining category, but it requires an insurance company to assess the methodology to tag NSPBO or remaining category based on

- likelihood of changes in assumptions which, if they occurred, would result in the contracts becoming onerous
- using information about estimates provided by the entity's internal reporting

Source: IFRS 17, Para. 19

Re-read this section and edit if needed

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What to Measure

On initial recognition, an entity shall measure a group of insurance contracts at the total of:

- the Fulfilment Cash Flows (FCF), which comprise:
  - estimates of future cash flows
  - an adjustment to reflect the time value of money and the financial risks related to the future cash flows, to the extent that the financial risks are not included in the estimates of the future cash flows, and
  - Risk Adjustment (RA) for non-financial risk
- the Contractual Service Margin (CSM)

Source: IFRS 17, Para. 32

We can re-write FCF in a more familiar term to be:

$$\mathrm{FCF} := E\left[\mathrm{PVCF}\right] + \mathrm{RA} = \mathrm{BEL} + \mathrm{RA} \tag{2}$$

where BEL stands for Best Estimate Liability. However we are going to discuss on, which future cash flow should be included in BEL calculation, how to choose the discount rate and how to calculate RA.

Contract Boundary I

Tomorrow's problems

Discount Rate

The discount rates applied to the estimates of the future cash flows described previously shall:

1. reflect the time value of money, the characteristics of the cash flows and the **liquidity characteristics** of the insurance contracts;
2. be consistent with observable current market prices (if any) for financial instruments with cash flows whose characteristics are consistent with those of the insurance contracts, in terms of, for example, timing, currency and liquidity; and
3. exclude the effect of factors that influence such observable market prices but do not affect the future cash flows of the insurance contracts.

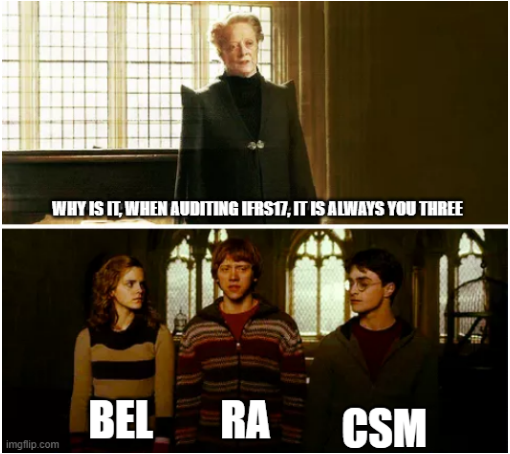
Source: IFRS 17, Para. 36

What interesting in the paragraph above is that we may use a lower or higher discount rate based on the liquidity characteristcs of the insurance contracts which means there is an additional assessment on how to determine how liquid is an insurance contract and the corresponding *buffer* one should add or substract from the current discount rate.

Risk Adjustment

CSM and LC

Will always be



Measurement Model

Explain on there are differents model for different type of insurance (GMM, PAA, and VFA). It is optional to use GMM or PAA but it is mandatory to use VFA if the insurance contract, pass the eligibility.

All model

VFA

VFA direct participating

PAA

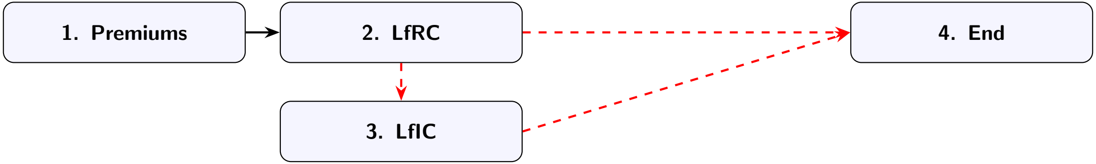
PAA short term

Bit of Bonus on Transisition Method

This is actually a set up to ending remarks. FVA, FRA, MRA

What happened to the premium so far

Based on the explanation so far, the premium paid to the insurer will directly become a LfRC then what happened ?



But can LfIC be calculated at initial recognition?

Remeasurement

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What is LfIC and how to calculate

IBNR

put my undergrad thesis a bit and a bit from exam 5

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Financial Disclosure and Statement Mapping

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Where does it show up in the financial statements?

- **Balance sheet:** insurance contract assets or liabilities (and reinsurance assets).
- **P&L:**
  - insurance service result (revenue, service expenses, onerous losses),
  - insurance finance income/expenses (unwind + discount-rate changes; presentation policy matters).
- **Disclosures:** reconciliations/roll-forwards tying opening to closing balances.

Remeasurement mapping (illustrative)

Driver	**Actuarial output**	**Accounting impact**
New claims information	Updated assumptions / triangles	Past service effect in service result; LfIC movement
Future-service assumption changes	Updated projection model	Unlock CSM (future service) and change in LfRC
Discount curve unwind / updates	Interest accretion / OCI option	Finance income/expense (P&L and/or OCI) and remeasurement
Onerous recognition / LC	Profit testing	Immediate loss in P&L; LC tracking in LfRC

Suggested reading (primary sources)

- International Accounting Standards Board (IASB). *IFRS 17 Insurance Contracts*. IFRS Foundation, issued 2017 (effective 1 January 2023).
- EY. *Applying IFRS 17: A closer look at the new Insurance Contracts Standard* ("EY IFRS 17 playbook" / guide).